

**Comparative Analysis of Modern Video Codecs: H.264, H.265, and AV1 Using Objective
and Perceptual Metrics**

Final Report

Xi Wang

CS4490Z Thesis

Department of Computer Science

Western University

2025/04/24

Project Supervisor: Prof. Umair Rehman, Department of Computer Science

Course Instructor: Prof. Nazim Madhavji, Department of Computer Science

Glossary

Term	Definition
Mean Opinion Score (MOS)	A subjective quality metric ranging from 1 (poor) to 5 (excellent) based on user ratings.
Video Multi-method Assessment Fusion (VMAF)	A perceptual video quality metric developed by Netflix that combines multiple metrics using machine learning.
Peak Signal-to-Noise Ratio (PSNR)	A traditional metric that compares the maximum signal power to the power of distortion or noise.
Kernel Inception Distance (KID)	A GAN-based image quality metric used here to compare frames between original and compressed videos.
AOMedia Video 1 (AV1)	An open, royalty-free video codec designed for high compression efficiency and quality.
H.264 / H.265 (AVC / HEVC)	Widely used video compression standards; H.265 offers better efficiency than H.264.
User Generated Content (UGC)	Videos created and uploaded by users, often found on platforms like YouTube or TikTok.

Structured Abstract

Background

As video streaming continues to dominate internet traffic, selecting the right video codec is critical for balancing quality, compression efficiency, and computational cost. This project evaluates three leading video codecs—H.264, H.265, and AV1—across a range of encoding parameters.

Objective

To systematically evaluate the trade-offs among compression time, output file size, and video quality across modern video codecs (H.264, H.265, and AV1), under various bitrate settings. The analysis incorporates both objective metrics (PSNR) and perceptual metrics (VMAF, KID, MOS) to provide a comprehensive comparison of codec performance.

Methodology

Videos from the YouTube UGC Dataset were encoded using each codec across multiple bitrate levels and encoding presets. Quality metrics (VMAF, PSNR, KID) were extracted from compressed videos. MOS values from the original videos were also used for correlation analysis.

Results

AV1 produced the best quality-to-size ratio but at a high computational cost. H.265 offered strong performance at moderate cost. KID and VMAF correlated better with MOS than PSNR. Higher bitrates improved quality but increased size and time.

Conclusion

For optimal trade-offs, H.265 with a slow preset at 2000k bitrate provides a strong balance. KID and VMAF are better predictors of perceptual quality than PSNR.

Table of Contents

1. Introduction
2. Background and Related Work 2.1 Video Codecs 2.2 Video Quality Metrics 2.3 Empirical Evaluation in Video Compression 2.4 Research Gap
3. Research Objectives
4. Methodology
5. Results
6. Discussion
7. Conclusions
8. Future Work and Lessons Learnt
9. Acknowledgments
10. Appendix
11. References

1. Introduction

The explosion of video streaming platforms has drastically increased demand for efficient video compression technologies. With video content accounting for over 80% of global internet traffic, even small improvements in codec efficiency can translate into large bandwidth and storage savings. This thesis investigates three widely used codecs—H.264 (ITU-T, 2010), H.265 (Sullivan & Wiegand, 2005), and the newer AV1 (Alliance for Open Media, 2025)—in terms of compression quality, speed, and file size.

The key research questions are:

- RQ1: Which codec provides the best perceptual video quality at a given bitrate?
- RQ2: What is the trade-off between encoding speed and quality across codecs?
- RQ3: How do objective quality metrics correlate with subjective MOS?

The study's novelty lies in its use of both perceptual metrics (VMAF, KID) and classical metrics (PSNR), and their correlation with MOS from the UGC dataset. The results can guide codec selection for video streaming applications.

2. Background and Related Work

2.1 Video Codecs

H.264 is a legacy standard known for broad hardware support. H.265 improves compression efficiency but increases computational cost. AV1, designed by the Alliance for Open Media, achieves superior compression at the expense of speed.

2.2 Video Quality Metrics

VMAF (Netflix, 2016) combines multiple perceptual models to predict user-perceived quality. PSNR measures pixel-level fidelity but poorly correlates with human perception. KID (PyTorch-Metrics, 2025) leverages deep neural embeddings and correlates better with perceptual similarity.

2.3 Empirical Evaluation in Video Compression

Past studies (Ruediger, Garth, Hagen, & Leitte, 2020) have compared codecs using PSNR/VMAF but lack integration with MOS benchmarks and KID analysis.

2.4 Research Gap

Prior works focus on one or two metrics or lack AV1 benchmarking. This study fills the gap by:

- Using real-world UGC content with MOS.
- Integrating VMAF, PSNR, and KID.
- Evaluating encoding time and file size.

3. Research Objectives

3.1 Research Goal and Objectives

Goal: To compare the performance of H.264, H.265, and AV1 codecs using objective and perceptual metrics to identify the most efficient codec for streaming platforms.

Research Question: How do these codecs perform in terms of quality, compression efficiency, and computational complexity under varying encoding parameters?

Objectives:

- O1:** Measure and compare the visual quality of AV1, H.264, and H.265 compressed videos.
- O2:** Quantify the compression time and file size of each codec under varying parameters.
- O3:** Correlate objective metrics (VMAF, PSNR, KID) with subjective scores (MOS).
- O4:** Analyze trade-offs between quality and efficiency across different content types.

3.2 Research Approach and Methodology

1. Dataset:

- a. Source: YouTube UGC Dataset.
- b. Content Types: Animation
- c. Categories:
 - i. High-motion.
 - ii. Slow-motion
 - iii. Mixed complexity.
- d. Resolutions: 360p, 480p, 720p, 1080p, 2160p.
- e. MOS Ratings File: original_videos_MOS_for_YouTube_UGC_dataset.xlsx

vid	category	resolution	# subjects	MOS full	MOS chunk00	MOS chunk05	MOS chunk10	std full	std chunk00	std chunk05	std chunk10
Animation_108C	Animation	1080	125	3.649	3.604	3.721	3.338	0.468	0.574	0.61	0.618
Animation_108C	Animation	1080	146	3.901	3.757	3.671	3.728	0.871	0.734	0.733	0.738
Animation_108C	Animation	1080	127	3.94	3.831	4.01	4.031	0.529	0.527	0.626	0.692
Animation_108C	Animation	1080	135	4.245	4.329	4.249	4.249	0.587	0.509	0.494	0.611
Animation_108C	Animation	1080	139	4.158	4.177	3.815	4.02	0.579	0.587	0.612	0.683
Animation_108C	Animation	1080	136	4.189	4.138	4.112	4.224	0.622	0.606	0.594	0.509
Animation_108C	Animation	1080	115	3.791	3.719	3.733	3.62	0.486	0.526	0.481	0.588
Animation_108C	Animation	1080	105	3.346	3.456	3.341	3.324	0.82	0.578	0.721	0.64

Figure 1: Example Data in original_videos_MOS_for_YouTube_UGC_dataset.xlsx

2. Compute Environment Setup:

- a. Tools: Google Cloud Platform (GCP) Virtual Machine (VM)
- b. Specifications:
 - i. Machine Type: e2-custom-6-16384 (6 vCPUs, 16 GB memory).
 - ii. CPU Platform: AMD Rome, x86/64 architecture
 - iii. Operating System: Debian-based Linux (default GCP image)
- c. Disk Configuration:

- i. Disk 1 (10 GB): Scripting, general computation (e.g., batch automation, evaluation logic)
- ii. Disk 2 (100 GB): Storage for encoded/compressed videos (.mp4 format)
- iii. Disk 3 (100 GB): Storage for original animation videos (.mp4 format)
- iv. Disk 4 (290 GB): Temporary storage for .y4m videos used in VMAF/PSNR evaluation

Note: Due to the 500 GB total storage limit on GCP free-tier and quota constraints, disk space was a major limiting factor in large-scale .y4m and KID evaluation experiments. Strategic sampling methods were employed to mitigate storage bottlenecks.

3. Compression Setup:

- a. Tools: FFmpeg for encoding/decoding.
- b. Parameters:
 - i. Codecs: H.264, H.265, AV1.
 - ii. Bitrates: 500 kbps, 1000 kbps, 2000 kbps, 4000 kbps.
 - iii. Presets: Ultrafast, Fast, Medium, Slow, Very Slow.
 - iv. Keyframe Intervals: 30, 60, 120 frames.

Note: Each combination generated a unique file. Compression logs were saved. VMAF and PSNR were extracted using Netflix's VMAF library(Netflix, 2016). KID was computed using TensorFlow-GAN (GAN Metrics, n.d.) on 50 extracted frames per video.

4. Video Quality Assessment (VQA):

- a. Metrics:
 - i. Objective: VMAF, PSNR, KID.
 - ii. Subjective: MOS.
- b. Tools:
 - i. Conversion to Y4M: `convert_encoded_to_y4m.sh`
 - ii. VMAF/PSNR scores Evaluation: `evaluate_codec_y4m.sh` using Netflix VMAF and FFmpeg PSNR
- c. Sample Size: 408 videos (due to storage limits)

5. KID Metric Computation:

- a. Sampling: 300 encoded videos covering diverse configurations (specified in Challenges)
- b. Tools:
 - i. Frame Extraction Scripts: `extract_frames_commands.sh`, `extract_compressed_frames_commands.sh`
 - ii. Evaluation Script: `updated_batch_kid_evaluation.py` (50 frames/video), TensorFlow Version: 2.12.0
TensorFlow Probability Version: 0.19.0
 - iii. Challenges: Out of Memory (OOM) errors due to large memory use of TensorFlow; final dataset = 180 entries

6. Data Collection:

- For each test case (codec/bitrate/preset combination), collected the following data:
 - Codec type (H.265, H.264, AV1).
 - Bitrate and encoding settings.
 - Output file size.
 - Compression time.
 - VMAF and PSNR scores.
 - Observations on any visible artifacts or quality degradation.
- Data Files Obtained:
 - Merged Data File (included all the scores in the below .xlsx files):

Merged_Encoding_Log.xlsx

filename	codec	bitrate	preset	keyint	output_file_size_bytes	compression_time_seconds	compressed	vmaf	psnr	kid_mean
Animation_1080P-05f8_crf_10_ss_00_t_20.0	libx264	500k	ultrafast	30	923210	15.49105	Animation_1080P-05f8_crf_10_ss_00_t_20.0_libx264_500k_ultrafast_keyint30	36.428924	23.576002	0
Animation_1080P-05f8_crf_10_ss_00_t_20.0	libx264	500k	ultrafast	60	857521	14.63705	Animation_1080P-05f8_crf_10_ss_00_t_20.0_libx264_500k_ultrafast_keyint60	36.637649	23.68469	0
Animation_1080P-05f8_crf_10_ss_00_t_20.0	libx264	500k	ultrafast	120	840577	14.7256	Animation_1080P-05f8_crf_10_ss_00_t_20.0_libx264_500k_ultrafast_keyint120	35.474322	23.637464	0
Animation_1080P-05f8_crf_10_ss_00_t_20.0	libx264	500k	fast	30	851361	33.1849	Animation_1080P-05f8_crf_10_ss_00_t_20.0_libx264_500k_fast_keyint30	34.59655	24.962519	0
Animation_1080P-05f8_crf_10_ss_00_t_20.0	libx264	500k	fast	60	833490	32.6343	Animation_1080P-05f8_crf_10_ss_00_t_20.0_libx264_500k_fast_keyint60	33.273995	24.868099	0

Figure 2: Example Data in Merged_Encoding_Log.xlsx

- Compression Data: encoding_log.xlsx.

filename	codec	bitrate	preset	keyint	output_file_size	compression_time_seconds
Animation_1080P-05f8_crf_10_ss_00_t_20.0	libx264	500k	ultrafast	30	923210	17.3196
Animation_1080P-05f8_crf_10_ss_00_t_20.0	libx264	500k	ultrafast	60	857521	16.7267
Animation_1080P-05f8_crf_10_ss_00_t_20.0	libx264	500k	ultrafast	120	840577	16.4574
Animation_1080P-05f8_crf_10_ss_00_t_20.0	libx264	500k	fast	30	851361	37.0384
Animation_1080P-05f8_crf_10_ss_00_t_20.0	libx264	500k	fast	60	833490	36.4202
Animation_1080P-05f8_crf_10_ss_00_t_20.0	libx264	500k	fast	120	828756	32.3052
Animation_1080P-05f8_crf_10_ss_00_t_20.0	libx264	500k	medium	30	872306	19.761
Animation_1080P-05f8_crf_10_ss_00_t_20.0	libx264	500k	medium	60	854753	19.1615

Figure 3: Example Data in encoding_log.xlsx

- VMAF/PSNR Scores Evaluation: evaluation_log.xlsx.

filename	codec	bitrate	preset	keyint	vmaf	psnr
Animation_1080P-05f8_crf_10_ss_00_t_20.0_libx264_1000k_fast_keyint120	h264	1000k	fast	120	40.505594	25.690842
Animation_1080P-05f8_crf_10_ss_00_t_20.0_libx264_1000k_fast_keyint30	h264	1000k	fast	30	43.167454	25.766418
Animation_1080P-05f8_crf_10_ss_00_t_20.0_libx264_1000k_fast_keyint60	h264	1000k	fast	60	41.878251	25.767856
Animation_1080P-05f8_crf_10_ss_00_t_20.0_libx264_1000k_medium_keyint120	h264	1000k	medium	120	47.469656	25.817012
Animation_1080P-05f8_crf_10_ss_00_t_20.0_libx264_1000k_medium_keyint30	h264	1000k	medium	30	49.584817	25.802187
Animation_1080P-05f8_crf_10_ss_00_t_20.0_libx264_1000k_medium_keyint60	h264	1000k	medium	60	49.328285	25.871221
Animation_1080P-05f8_crf_10_ss_00_t_20.0_libx264_1000k_slow_keyint120	h264	1000k	slow	120	49.1987	25.845305

Figure 4: Example Data in evaluation_log.xlsx

- KID Scores Evaluation: kid_scores_50_frames.xlsx.

original	compressed	kid_mean	kid_std
Animation_720P-3adc_crf_10_ss_00_t_20.0	Animation_720P-3adc_crf_10_ss_00_t_20.0_libx265_500k_fast_keyint120	0.003035426	0
Animation_720P-3adc_crf_10_ss_00_t_20.0	Animation_720P-3adc_crf_10_ss_00_t_20.0_libx265_2000k_slow_keyint120	0	0
Animation_720P-3adc_crf_10_ss_00_t_20.0	Animation_720P-3adc_crf_10_ss_00_t_20.0_libx264_500k_fast_keyint120	0.002923727	0
Animation_720P-3adc_crf_10_ss_00_t_20.0	Animation_720P-3adc_crf_10_ss_00_t_20.0_libx265_500k_ultrafast_keyint30	0.010648489	0
Animation_720P-3adc_crf_10_ss_00_t_20.0	Animation_720P-3adc_crf_10_ss_00_t_20.0_libx265_4000k_slow_keyint30	0	0
Animation_720P-3adc_crf_10_ss_00_t_20.0	Animation_720P-3adc_crf_10_ss_00_t_20.0_libx265_4000k_ultrafast_keyint60	0	0
Animation_720P-3adc_crf_10_ss_00_t_20.0	Animation_720P-3adc_crf_10_ss_00_t_20.0_libx265_1000k_slow_keyint60	0.000686646	0
Animation_1080P-2fbc_crf_10_ss_00_t_20.0	Animation_1080P-2fbc_crf_10_ss_00_t_20.0_libx265_1000k_fast_keyint60	0.122492552	0
Animation_1080P-2fbc_crf_10_ss_00_t_20.0	Animation_1080P-2fbc_crf_10_ss_00_t_20.0_libx265_1000k_fast_keyint30	0.124633312	0
Animation_1080P-2fbc_crf_10_ss_00_t_20.0	Animation_1080P-2fbc_crf_10_ss_00_t_20.0_libx264_500k_slow_keyint120	0.201170921	0
Animation_1080P-2fbc_crf_10_ss_00_t_20.0	Animation_1080P-2fbc_crf_10_ss_00_t_20.0_libx264_1000k_medium_keyint120	0.120267391	0

Figure 5: Example Data in kid_scores_50_frames.xlsx

7. Analysis:

- **Bitrate vs. Video Quality:** Created visual plots to explore the relationship between bitrate and video quality (VMAF/PSNR) for each codec, highlighting differences in performance.
- **Compression Time vs. Video Quality:** Evaluated the trade-offs between compression speed and the resulting video quality across all codecs, providing insights into efficiency.
- **File Size vs. Video Quality:** Compared the file sizes of compressed videos and identified the codec capable of achieving the smallest size without significant quality degradation.
- **Computational Load:** Examined the computational efficiency of each codec, considering time, memory, and processing power requirements during encoding and decoding.

Validation:

Results validated by correlating subjective MOS scores with objective metrics to ensure alignment with user-perceived quality.

4. Results

4.1 Bitrate vs. Video Quality

Across all codecs, VMAF and PSNR scores increased with bitrate. AV1 consistently achieved higher VMAF at lower bitrates. KID values decreased with bitrate, indicating better similarity (see Figure 6).

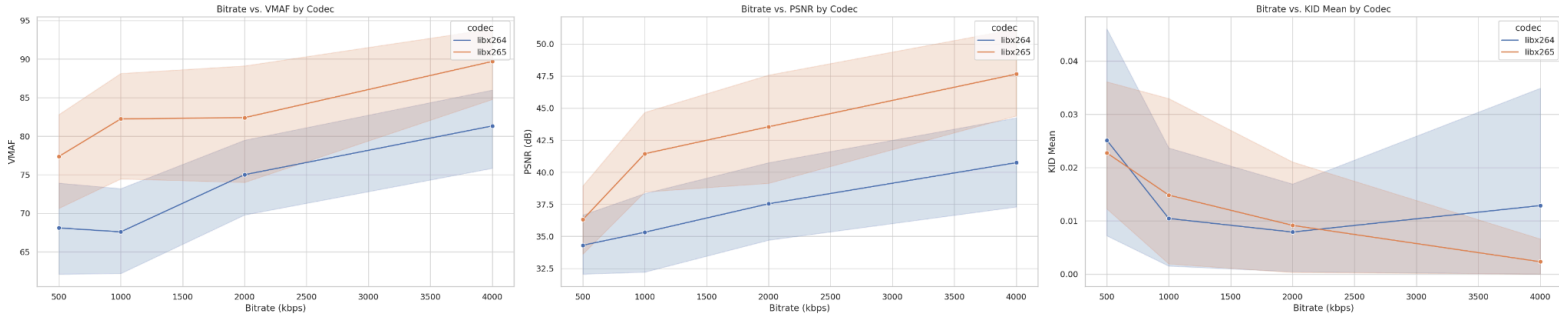


Figure 6: Bitrate vs. Video Quality (VMAF/PSNR/KID)

4.2 Compression Time vs. Video Quality

AV1 was 3–5 $\times$ slower than H.265 and 10 $\times$ slower than H.264. H.265 slow preset achieved similar quality to AV1 at 30–50% of the encoding time (see Figure 7).

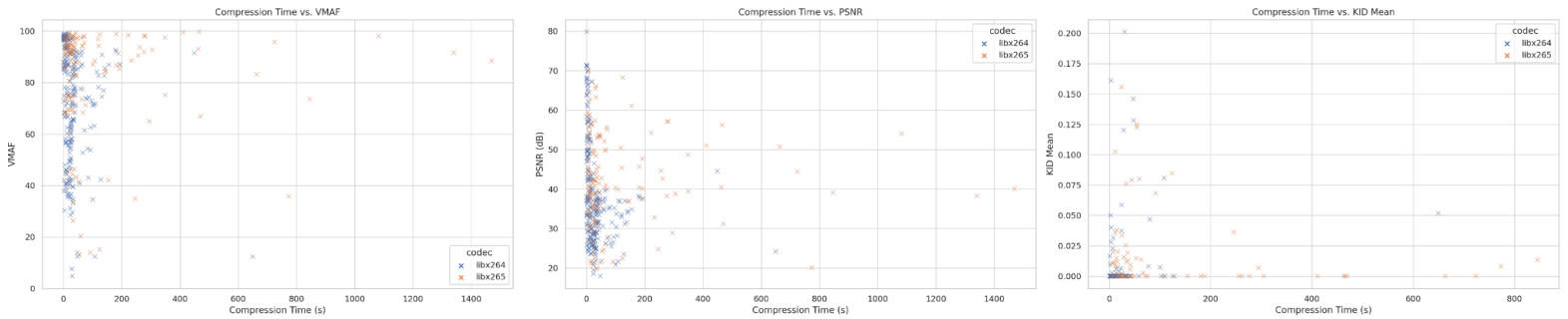


Figure 7: Compression Time vs. Video Quality (VMAF/PSNR/KID)

4.3 File Size vs. Video Quality

AV1 offered the smallest size for the same quality. H.265 struck a balance between H.264 speed and AV1 compression (see Figure 8).

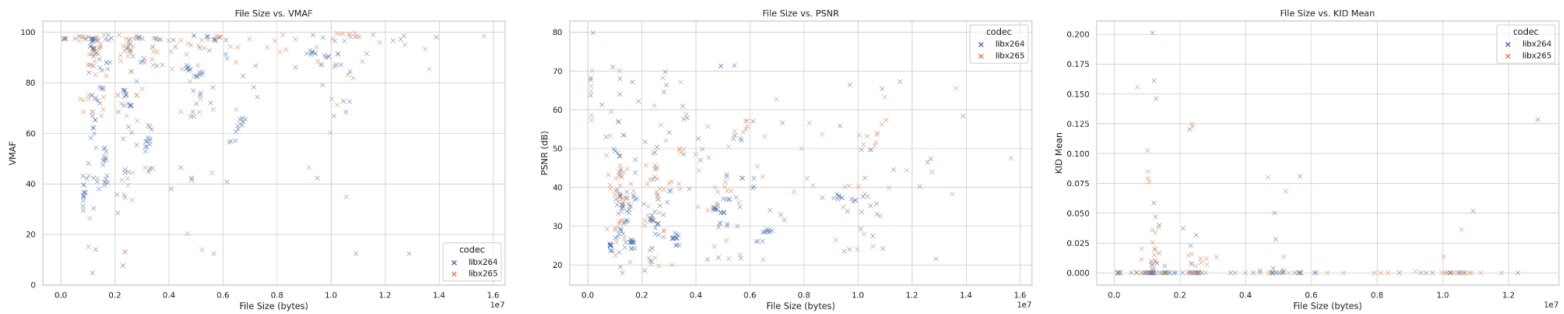


Figure 8: File Size vs. Video Quality (VMAF/PSNR/KID)

4.4 MOS vs. Metrics

VMAF and KID had stronger correlation with MOS (Netflix, 2016; Lightning AI, 2025). PSNR showed weaker correlation (see Figures 9).

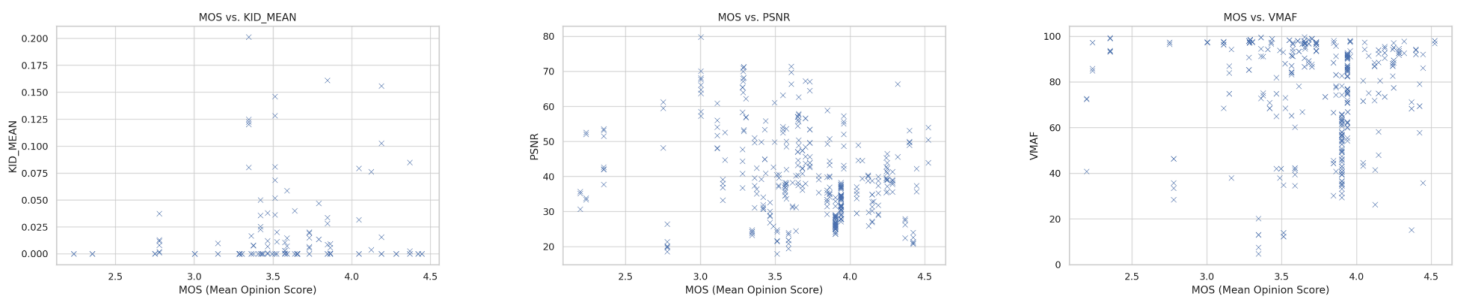


Figure 9: MOS vs. Other Metrics (VMAF/PSNR/KID)

4.5 Compression Time Comparison of different codec

The graph (see Figures 10) clearly highlights the computational trade-offs associated with more advanced codecs. H.264 remains the most efficient in terms of encoding speed, while AV1 introduces significant time overhead for the benefit of superior compression and quality. H.265 (libx265) strikes a reasonable balance between speed and efficiency, making it suitable for a wider range of real-world applications.

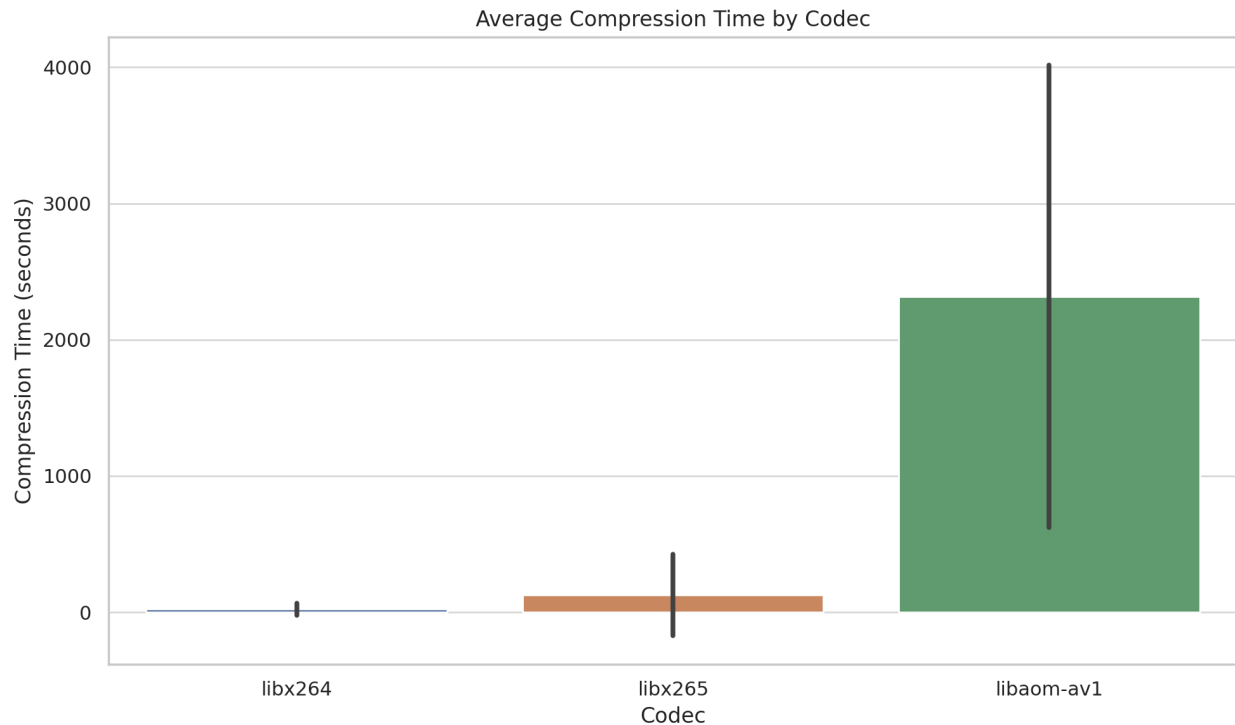


Figure 10: Average Compression Time by AV1, H264, H265 Codec

5. Discussion

Threats to Validity

Several factors may affect the validity of this study's results. First, the dataset used in the experiments—although representative of common animation sequences—was relatively limited in size and scope. This constraint may introduce bias, particularly in the Kernel Inception Distance (KID) scores, which are sensitive to the diversity and number of extracted frames. Additionally, variations in AV1 encoder implementations and configurations across platforms could result in significantly different performance outcomes. These platform-level differences may impact the generalizability of both compression time and quality metrics.

Implications

The findings of this study have meaningful implications for practitioners, especially streaming service providers. AV1 demonstrates superior performance in terms of compression efficiency and perceptual quality metrics (e.g., VMAF and KID), making it a compelling choice for long-term video storage and bandwidth-constrained delivery. However, the trade-off lies in its substantially longer encoding time, which may pose a bottleneck in real-time or large-scale encoding pipelines. In contrast, H.265 (HEVC) emerges as a more balanced alternative—it offers improved compression over H.264 while maintaining reasonable encoding speed, making it a practical middle-ground for many real-world use cases where both efficiency and throughput matter.

Limitations

VMAF, while a powerful perceptual quality metric, depends on a fixed reference video, which assumes frame-by-frame alignment and may not account for temporal distortions or non-reference-based scenarios. Furthermore, KID scores are influenced by both the number of frames extracted and the precision of the underlying GAN-based model used to compute distances (GAN Metrics, n.d.). Inconsistencies in frame sampling or model performance could affect metric stability and comparability across videos and codecs.

Generalisability

Despite the limitations, the experimental methodology used in this study can be extended to other video codecs or content types. Given a standardized metric pipeline—including consistent frame extraction, bitrate control, and evaluation using both objective and perceptual quality metrics—this framework can be applied to a wide range of video compression evaluations. It offers a reproducible and adaptable approach for future comparative studies across different codecs, content genres, or encoding environments.

6. Conclusions

This study conducted a comprehensive comparative analysis of three modern video codecs—AV1 (libaom-av1), H.264 (libx264), and H.265 (libx265)—using a combination of perceptual quality metrics and computational performance indicators. The goal was to evaluate the trade-offs between compression time, file size, and visual quality across different encoding strategies and bitrate settings, with a particular emphasis on the relevance of perceptual metrics in reflecting human viewing experience.

Key findings

- AV1 demonstrated the best quality-to-size efficiency, producing the smallest file sizes while preserving the highest perceptual video quality, as measured by VMAF and KID. This makes AV1 highly suitable for use cases where bandwidth is constrained and storage efficiency is critical, such as large-scale streaming or long-term archival.
- H.265 (HEVC) emerged as a strong compromise between compression efficiency and encoding speed. While not as compact as AV1 in terms of output file size, H.265 achieved comparable perceptual quality with significantly lower compression time, offering a practical solution for real-world encoding workflows that require both speed and quality.
- When comparing the effectiveness of video quality metrics, it was observed that VMAF and KID showed stronger alignment with subjective human ratings (MOS) than traditional pixel-based metrics such as PSNR. These findings highlight the limitations of PSNR in capturing the nuances of human visual perception and underscore the importance of adopting perceptual metrics in modern video quality assessment frameworks.

7. Future Work and Lessons Learnt

Future Work

While this study provided valuable insights into the comparative performance of modern video codecs, there remain several areas for further exploration and improvement. First, future work could incorporate real-time playback evaluations to assess how compression artifacts affect the actual viewing experience during live or interactive scenarios. This would complement the current frame-based analysis and help bridge the gap between metric-based evaluations and user-perceived performance. Second, the current Kernel Inception Distance (KID) analysis was limited to a relatively small number of extracted frames per video. Extending this evaluation to a larger sample—such as 100 or more uniformly sampled frames—could improve statistical reliability. Additionally, evaluating KID across multiple resolutions would provide a more nuanced understanding of how perceptual quality degrades or scales with resolution changes. Third, future experiments could explore hardware-accelerated encoding environments, such as using NVIDIA NVENC or Intel Quick Sync for H.264 and H.265, or dedicated AV1 hardware encoders. These tests would offer practical insights into codec performance in production settings, where GPU or ASIC-based encoding is common.

Lessons Learnt

Throughout the course of this study, several important technical and methodological lessons emerged. One key insight is that frame-level metrics such as PSNR may fail to capture meaningful visual differences, especially in complex or high-motion scenes. In contrast, perceptual models like VMAF and KID provided more reliable and human-aligned assessments of video quality, reinforcing the importance of perceptual metrics in modern codec evaluation. Another lesson is the importance of maintaining consistent and unified file naming conventions, particularly when dealing with large datasets and multi-step processing pipelines. Uniform naming across resolutions, codecs, and CRF settings proved essential for efficient log parsing, metric alignment, and post-processing. This lesson underscores the value of systematic data organization in large-scale video processing tasks.

8. Acknowledgments

Many thanks to the supervisor, whose guidance, support, and insightful feedback were instrumental throughout the development of this project. His expertise helped shape both the technical direction and overall structure of the research, and their encouragement was invaluable during critical phases of experimentation and writing.

The research was made possible through the use of several open-source tools and frameworks. TensorFlow-GAN was essential in implementing the Kernel Inception Distance (KID) metric, enabling robust perceptual quality assessment across compressed video frames. Despite computational constraints, this tool allowed for meaningful comparisons between codecs based on generative similarity. The Netflix VMAF framework also played a critical role by providing a perceptually grounded method for evaluating video quality, supporting the study's goal of aligning objective metrics with subjective human ratings.

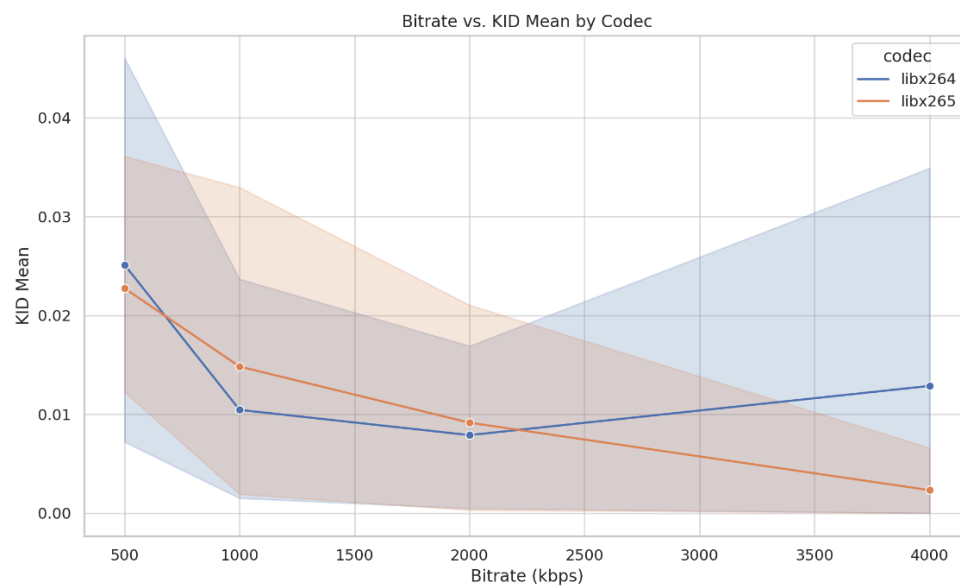
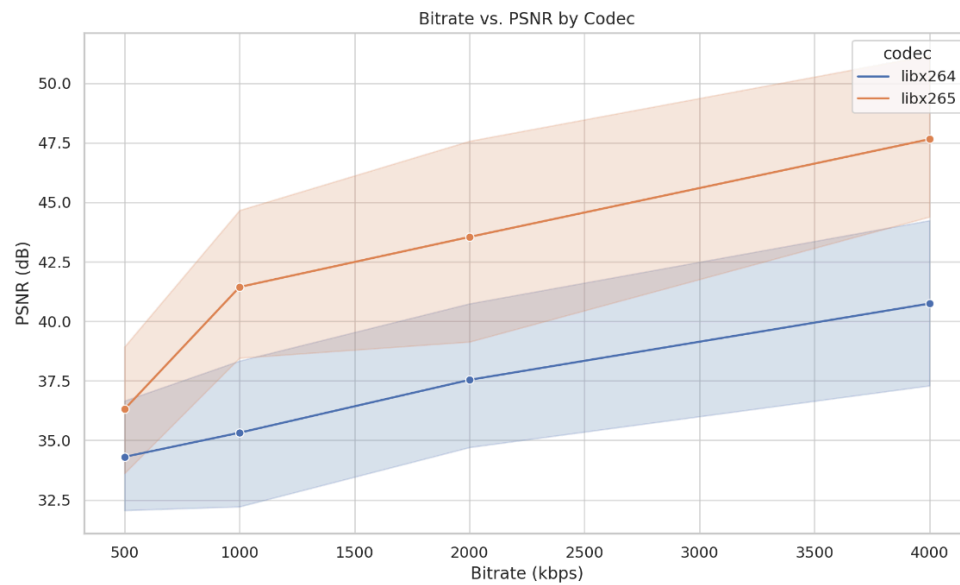
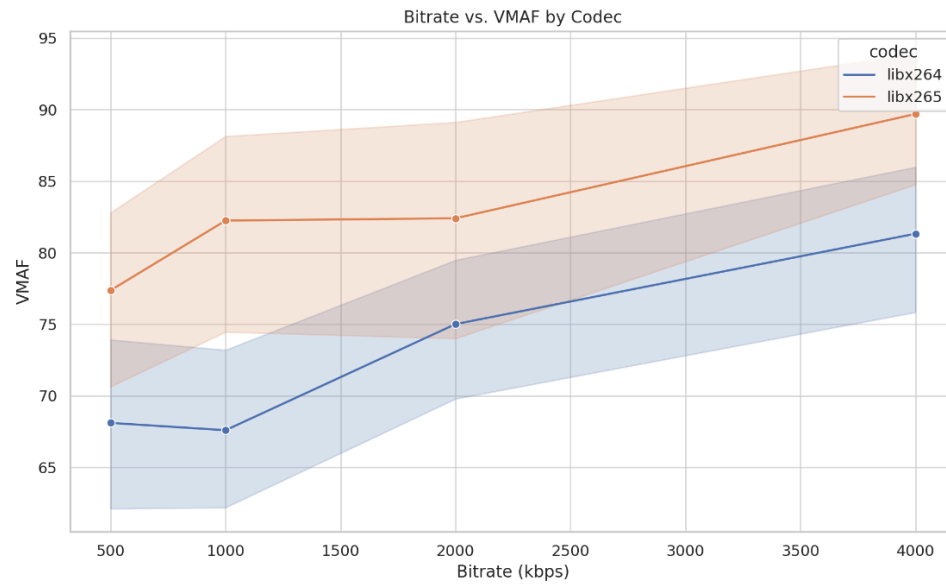
Gratitude is also extended to the creators of the YouTube UGC Dataset, which serves as a valuable source of diverse animation content. The inclusion of Mean Opinion Score (MOS) data was particularly important for validating metric performance against human perception.

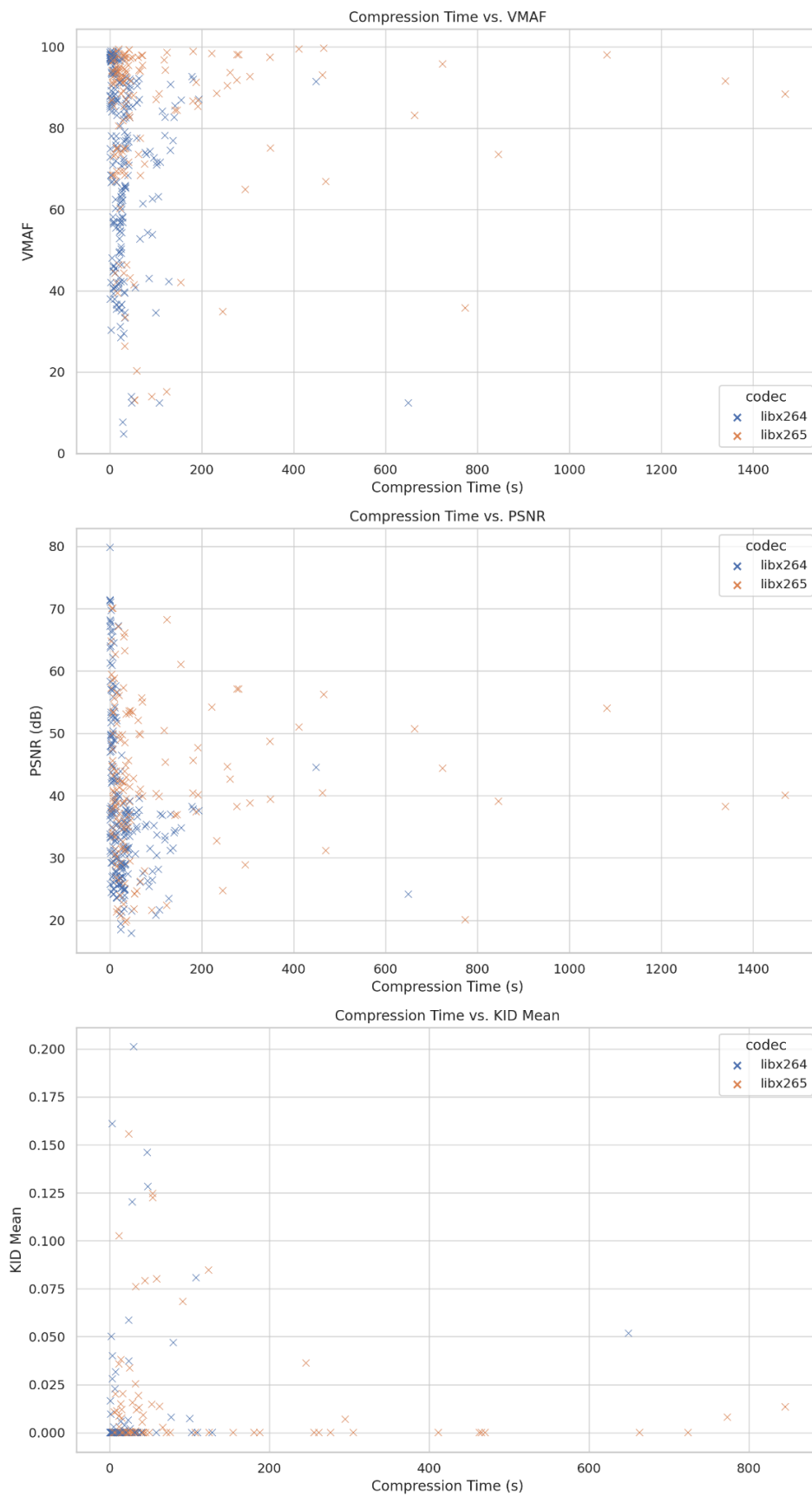
Appreciation is further extended to the broader open-source and academic communities whose tools—including FFmpeg, Google Cloud Platform (GCP) services, and supporting scripts—facilitated the data processing and experimentation workflows. These resources were fundamental to the study's ability to evaluate codec performance under varying encoding conditions and to meet the outlined research objectives.

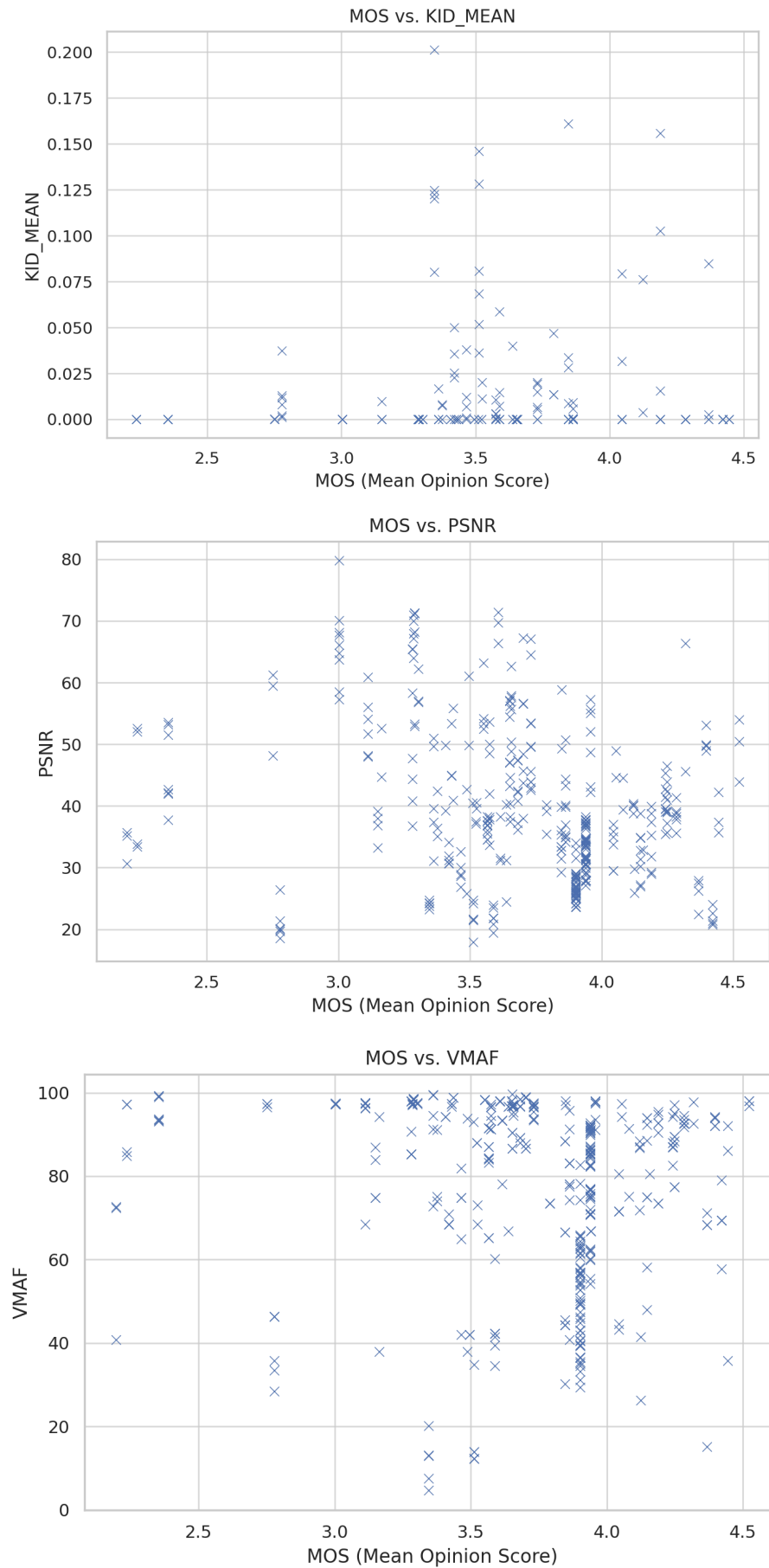
9. Appendix

See attached graphs for:

- Bitrate vs VMAF/PSNR/KID
- File Size vs VMAF/PSNR/KID
- Compression Time vs VMAF/PSNR/KID
- MOS Metric vs Other Metrics







References

Alliance for Open Media. (2025). AV1 Codec Overview. Retrieved from [official website].

<https://aomedia.org/specifications/av1>

GAN Metrics — Machine Learning Note. Retrieved from [official website]

https://machine-learning-note.readthedocs.io/en/latest/generative_models/metrics.html

ITU-T. (2010). Recommendation ITU-T H.264: Advanced Video Coding for Generic

Audiovisual Services. Retrieved from [official website].

Netflix. (2016). Towards a Perceptual Video Quality Metric. Retrieved from [Netflix Tech Blog].

<https://netflixtechblog.com/toward-a-practical-perceptual-video-quality-metric-653f208b9652>

Lightning AI (2025). PyTorch-Metrics: Kernel Inception Distance. Retrieved from [official website]

https://lightning.ai/docs/torchmetrics/stable/image/kernel_inception_distance.html

Ruediger, Garth, Hagen, & Leitte (2020). Evaluating Video Compression Techniques for Simulation Ensembles. <https://doi.org/10.4230/OASICS.iPMVM.2020.10>

Sullivan, G. J., & Wiegand, T. (2005). Video Compression—From Concepts to the H.264/AVC Standard. *Proceedings of the IEEE*, 93(1), 18-31.